

A partial answer to the expected $p = 2$ rate in Chen–Yousept

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Status

This packet records a partial positive answer to Remark 4.9 of Chen–Yousept [1]. The source paper obtains, in the weakest-data case of Theorem 4.6(iv), a rate which is suboptimal when the penalty exponent is $p = 2$, and writes that the expected rate is

$$\|a_{\alpha(\delta)}^\delta - a^\dagger\|_2^2 = O\left(\delta^{\frac{2s}{s+2}}\right).$$

The result below proves this expected rate under a stronger true-state multiplier assumption and an adapted Hilbert-scale smoothness assumption. The varying coefficients remain in the source paper’s original $L^{\mathfrak{p}}$ -bounded admissible class, with $\mathfrak{p} \geq 2$.

As a conclusion, different choices of Y -norms and estimates (4.12)-(4.14) lead to different convergence rates. The case (iv) with the weakest norm $Y = L^\tau(\Omega)$ is mostly relevant for applications since measurement in higher Sobolev norms is typically difficult to realize in practice.

Remark 4.9. *In the case of (iv) of Theorem (4.6) with $p = 2$, one may expect convergence rate $\|a_{\alpha(\delta)}^\delta - a^\dagger\|_2^2 = O(\delta^{\frac{2s}{s+2}})$. Thus, our convergence rates is suboptimal in this case, which may be caused by the treatment of the nonlinearity of the forward operator. It is an open question that we shall investigate as a future goal.*

4.3 Discussion of hypotheses in Theorem 4.6

In the following, we discuss the assumptions (b), (4.13) and (4.14) with more details.

Figure 1: Remark 4.9 in [1], page 23.

Extracted question

In the notation of [1, Theorem 4.6(iv)], for the Hilbert penalty $p = 2$, can one recover the expected rate

$$\|a_{\alpha(\delta)}^\delta - a^\dagger\|_2^2 = O\left(\delta^{\frac{2s}{s+2}}\right)?$$

Idea of the proof

The loss in the source paper comes from estimating the coefficient difference in $H_q^1(\Omega)^*$ and then interpolating an H^2 -bound for the state difference with the weak observation norm. This introduces a square root in the stability estimate before the variational-source-condition argument starts.

For $p = 2$ one can instead use the equation itself. Let

$$w := S(a) - S(a^\dagger), \quad h := a^\dagger - a.$$

If A_0 is the fixed Neumann realization of $-\operatorname{div}(\kappa \nabla \cdot) + 1$, then testing the difference equation with $\varphi \in D(A_0)$ gives

$$\int_{\Omega} h u^\dagger \varphi \, dx = \langle w, A_0 \varphi \rangle_{L^2} + \int_{\Omega} (a - 1) w \varphi \, dx.$$

Since $D(A_0) \subset H^2(\Omega) \subset L^\infty(\Omega)$ in dimensions two and three, the source paper's L^p -bound on a , with $p \geq 2$, is enough to control the nonlinear remainder by the L^2 -size of w . Thus h is controlled linearly in the dual of the adapted second-order space $u^\dagger D(A_0)$. A standard Hilbert-scale frequency splitting then gives exactly the exponent $2s/(s+2)$.

Setting

Let $n \in \{2, 3\}$. Let $\Omega \subset \mathbb{R}^n$ be a bounded $C^{1,1}$ domain and let $\kappa \in C^{0,1}(\overline{\Omega})$ satisfy

$$0 < \lambda_0 \leq \kappa(x) \leq \Lambda < \infty.$$

Assume that the admissible coefficients are those of the source paper:

$$\mathcal{D} := \{a \in L^p(\Omega) : \|a\|_{L^p} \leq M, \, 0 < \underline{a} \leq a(x) \text{ a.e.}\}, \quad p \geq 2.$$

For $a \in \mathcal{D}$, let $S(a)$ be the weak solution of the Neumann problem considered in [1, Section 4], with fixed measure data. Fix a true coefficient $a^\dagger \in \mathcal{D}$ and write $u^\dagger := S(a^\dagger)$. Assume

$$0 < \gamma \leq |u^\dagger| \leq \Gamma < \infty \quad \text{a.e. on } \Omega.$$

Let $H = L^2(\Omega)$, and let A_0 be the positive self-adjoint Neumann realization in H of

$$-\operatorname{div}(\kappa \nabla \cdot) + 1.$$

Thus

$$\langle A_0 \varphi, \psi \rangle_H = \int_{\Omega} \kappa \nabla \varphi \cdot \nabla \psi + \varphi \psi \, dx, \quad \varphi \in D(A_0), \, \psi \in H^1(\Omega).$$

We use the standard H^2 -regularity $D(A_0) \hookrightarrow H^2(\Omega)$; for $n \leq 3$, this gives $D(A_0) \hookrightarrow L^\infty(\Omega)$. Define the adapted second-order space

$$X_2 := u^\dagger D(A_0), \quad \|g\|_{X_2} := \|A_0(g/u^\dagger)\|_2.$$

Since multiplication by $u^\dagger$ is boundedly invertible on H , X_2 is densely and continuously embedded in H . Let $(X_t)_{-2 \leq t \leq 2}$ be the Hilbert scale generated by $X_2 \hookrightarrow H$, and write X_{-2} for the dual of X_2 with pivot H .

Lemma 1 (Linear adapted X_{-2} -stability). *There is a constant C , depending only on the fixed ellipticity and admissible-class bounds, such that for every $a \in \mathcal{D}$,*

$$\|a^\dagger - a\|_{X_{-2}} \leq C \|S(a) - S(a^\dagger)\|_2.$$

Proof. Put $h = a^\dagger - a$ and $w = S(a) - S(a^\dagger)$. For $g \in X_2$, write $g = u^\dagger \varphi$ with $\varphi \in D(A_0)$. Subtracting the weak equations for $S(a)$ and $S(a^\dagger)$, tested against φ , gives

$$\int_{\Omega} h u^\dagger \varphi \, dx = \int_{\Omega} \kappa \nabla w \cdot \nabla \varphi + a w \varphi \, dx.$$

Adding and subtracting $\int_{\Omega} w\varphi dx$ on the right side yields

$$\int_{\Omega} h g dx = \langle w, A_0\varphi \rangle_H + \int_{\Omega} (a-1)w\varphi dx.$$

Since $\mathbf{p} \geq 2$, the conjugate exponent $\mathbf{p}' \leq 2$. By boundedness of Ω , $\|w\|_{\mathbf{p}'} \leq C\|w\|_2$. Since $\|a\|_{L^{\mathbf{p}}} \leq M$ and $D(A_0) \hookrightarrow L^{\infty}(\Omega)$, we get

$$\left| \int_{\Omega} (a-1)w\varphi dx \right| \leq C(1+M)\|w\|_2\|\varphi\|_{\infty} \leq C(1+M)\|w\|_2\|A_0\varphi\|_2.$$

The first term satisfies

$$|\langle w, A_0\varphi \rangle_H| \leq \|w\|_2\|A_0\varphi\|_2.$$

Combining the two estimates and using $\|g\|_{X_2} = \|A_0\varphi\|_2$ proves

$$\left| \int_{\Omega} h g dx \right| \leq C\|w\|_2\|g\|_{X_2}.$$

Taking the supremum over $\|g\|_{X_2} \leq 1$ proves the claim. $\square$

Lemma 2 (Hilbert-scale variational inequality). *Let $0 < s \leq 2$ and $f^{\dagger} := a^{\dagger} - a^* \in X_s$. For every $\beta > 0$ there is a constant $C_{\beta} > 0$ such that, for all $a \in \mathcal{D}$,*

$$\langle a^{\dagger} - a, f^{\dagger} \rangle_H \leq \frac{\beta}{2}\|a - a^{\dagger}\|_2^2 + C_{\beta}\|S(a) - S(a^{\dagger})\|_2^{\frac{2s}{s+2}}.$$

Proof. Let $G \geq I$ be the positive self-adjoint scale generator, so that $X_t = D(G^{t/2})$ for $t \geq 0$ and $\|z\|_{X_{-2}} = \|G^{-1}z\|_H$. Let E_{λ} be the spectral resolution of G , and set $P_R = E_G([1, R])$ for $R \geq 1$. For $h := a^{\dagger} - a$,

$$\langle h, f^{\dagger} \rangle = \langle h, (I - P_R)f^{\dagger} \rangle + \langle h, P_Rf^{\dagger} \rangle.$$

The high-frequency term satisfies, by Young's inequality,

$$|\langle h, (I - P_R)f^{\dagger} \rangle| \leq \|h\|_2 R^{-s/2} \|f^{\dagger}\|_{X_s} \leq \frac{\beta}{2}\|h\|_2^2 + C_{\beta}R^{-s}\|f^{\dagger}\|_{X_s}^2.$$

The low-frequency term is bounded by

$$|\langle h, P_Rf^{\dagger} \rangle| \leq \|h\|_{X_{-2}}\|P_Rf^{\dagger}\|_{X_2} \leq R^{(2-s)/2}\|h\|_{X_{-2}}\|f^{\dagger}\|_{X_s}.$$

Using Lemma 1, the total upper bound is

$$\frac{\beta}{2}\|h\|_2^2 + C_{\beta}R^{-s} + CR^{(2-s)/2}\|S(a) - S(a^{\dagger})\|_2.$$

Optimizing in R gives

$$R \simeq \|S(a) - S(a^{\dagger})\|_2^{-2/(s+2)}$$

when the data discrepancy is small; for large discrepancies the constant can be enlarged. Substitution yields the exponent $2s/(s+2)$. $\square$

Theorem 1 (Expected $p = 2$ rate in the adapted scale). *Assume the setting above, let $0 < s \leq 2$, and suppose $a^{\dagger} - a^* \in X_s$. Then the L^2 -penalty Tikhonov regularization problem satisfies a variational source condition with index*

$$\Psi(t) = Ct^{\frac{2s}{s+2}}.$$

Consequently, for the parameter choice used in [1, Corollary 3.2],

$$\|a_{\alpha(\delta)}^{\delta} - a^{\dagger}\|_2^2 = O\left(\delta^{\frac{2s}{s+2}}\right).$$

The same conclusion holds for data norms $Y = L^{\tau}(\Omega)$, $\tau \geq 2$, because Ω is bounded and $\|v\|_2 \leq C\|v\|_{\tau}$.

Proof. The variational source condition is Lemma 2. The convergence-rate claim is the standard implication from a variational source condition to Tikhonov rates, applied exactly as in [1, Corollary 3.2]. $\square$

Verification and limitations

The proof is complete for the stated strengthened setting. Compared with the previous partial version, Lemma 1 no longer uses a uniform L^∞ -bound on the varying admissible coefficients; the source paper’s L^p -bound with $p \geq 2$ is enough. The result still does not settle the full source-paper formulation, because the smoothness assumption is in the adapted scale X_s generated by $u^\dagger D(A_0)$, rather than directly in the classical Triebel–Lizorkin/Sobolev scale of the source paper. In smooth positive situations, this adapted scale is the natural Neumann elliptic Hilbert scale and is equivalent to the corresponding Sobolev scale after multiplication by $u^\dagger$.

Bounded literature checks were performed for the exact source title, arXiv id, the phrase “ $2s/(s+2)$ ”, and combinations of “unbounded diffusion”, “measure data”, “variational source condition”, and “Tikhonov”. No later paper explicitly resolving Remark 4.9 was found in those searches. The Hilbert-scale splitting itself is standard; the new point recorded here is the short adapted X_{-2} stability estimate for this nonlinear coefficient map, now with only the source paper’s L^p -control of the varying coefficient.

Human-review recommendation

First check Lemma 1; it is the only problem-specific step. If accepted, the remaining Hilbert-scale argument is standard. A useful next step would be to translate X_s into the exact Littlewood–Paley notation of [1] under the weakest true-state multiplier assumptions.

References

- [1] D.-H. Chen and I. Yousept, *Variational source conditions in L^p -spaces*, arXiv:2007.13474, 2020.
- [2] B. Hofmann and P. Mathé, *Parameter choice in Banach space regularization under variational inequalities*, Inverse Problems 28 (2012), 104006.